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Relevant sampling in a reproducing kernel subspace of Orlicz space

Shivam Bajpeyi

Department of Mathematics
Sardar Vallabhbhai National Institute of Technology
Surat 395007, India
shivambajpai1010@gmail.com

Dhiraj Patel* and S. Sivananthan†

Department of Mathematics
Indian Institute of Technology Delhi
New Delhi 110016, India
**dpatel.iitd@gmail.com*
†*siva@maths.iitd.ac.in*

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In this paper, we aim to provide a general paradigm for dealing with the sampling and random sampling problem in a reproducing kernel subspace of Orlicz space $L^{\phi}(\mathbb{R}^n)$. We consider the function space V as the image of an idempotent integral operator on $L^{\phi}(\mathbb{R}^n)$, where the integral kernel satisfies certain off-diagonal decay and regularity conditions. The model example of such reproducing kernel subspace of $L^{\phi}(\mathbb{R}^n)$ includes the finitely generated shift-invariant space and signal space with a finite rate of innovation. We show that a signal in V can be stably reconstructed from its samples at distinct points separated by a sufficiently small gap. Next, we deduce that the random sampling inequality holds with a high probability for the class of functions in V concentrated on a cube $C_R = [-\frac{R}{2}, \frac{R}{2}]^n$, when the samples collected at i.i.d. random points are drawn on C_R of order $\mathcal{O}(R^n \log R^n)$.

Keywords: Random sampling inequality; Orlicz space; reproducing kernel space; idempotent operator.

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1. Introduction

Sampling and reconstruction of any signal is an essential part of communication theory. The problem of sampling for a function space V is mainly concerned with finding a discrete sample set X under which any signal $f \in V$ can be stably reconstructed by employing its sample values $\{f(x_j) : x_j \in X\}$ without losing the original information. The set S is known as the stable set of sampling for V. A significant

*Corresponding author.

result in this direction was given by Whittaker–Kotelnikov–Shannon (see [8]), which asserts that any signal $f : \mathbb{R} \to \mathbb{C}$ that is band-limited to $[-\pi w, \pi w]$ for some $w > 0$, i.e. the Fourier transform of f is compactly supported in $[-\pi w, \pi w]$, can be completely reconstructed from its uniform sample values $(f(k/w))_{k\in\mathbb{Z}}$ by the formula

$$f(x) = \sum_{k\in\mathbb{Z}} f\left(\frac{k}{w}\right) \frac{\sin \pi(wx-k)}{\pi(wx-k)}, \quad x \in \mathbb{R}.$$

Some of the well-known developments related to the WKS sampling formula can be observed in [7, 13, 22, 25, 34, 35, 47]. It is one of the most significant results in communication theory and signal analysis. However, its applications are restricted since the signals are not necessarily band-limited in nature, as shown in [8, 45]. This inspired several researchers to study the sampling problem in shift-invariant spaces; for more information, we refer to [1–3, 19, 44].

Nashed and Sun [32, 33] investigated the problem of sampling for the image space V of an idempotent integral operator on $L^p(\mathbb{R}^n)$. The model space includes the space of spline functions and the shift-invariant space. In particular, they showed that under certain assumptions on the integral kernel, a relatively separated subset Γ of $\mathbb{R}^n$ is a stable sample set for V that satisfies the following sampling inequality:

$$A\|f\|^p_{L^p(\mathbb{R}^n)} \leq \sum_{\gamma\in\Gamma} |f(\gamma)|^p \leq B\|f\|^p_{L^p(\mathbb{R}^n)} \quad \forall f \in V,$$

where A and B are some positive constants.

In this paper, we will study the problem of sampling in a general setting of function spaces known as Orlicz space, denoted by $L^\phi(\mathbb{R}^n)$, where ϕ is an Orlicz function (see Sec. 2). The space $L^\phi(\Omega)$ is a Banach space containing all measurable functions on $\Omega \subseteq \mathbb{R}^n$ with the norm (also known as Luxemburg norm)

$$\|f\|_{L^\phi(\Omega)} := \inf\left\{\lambda > 0 : \int_\Omega \phi\left(\frac{|f(x)|}{\lambda}\right) dx \leq 1\right\}.$$

For $\phi(x) = x^p$ for $x \geq 0, 1 \leq p < \infty$, the corresponding Orlicz space will be reduced to the classical L^p-spaces.

The Orlicz spaces were first extensively discussed in [24], and subsequently, many researchers further investigated their characteristics and potential applications in various fields. Rao and Ren [41] presented the concrete theory of Orlicz spaces and their properties along with some related topics such as compact sets, the product of Orlicz spaces, topological structures, etc. Moreover, some fundamental applications in Fourier analysis, stochastic analysis, and nonlinear PDE have been discussed rigorously in [42] in the context of Orlicz space. For more information on the underlying theory of Orlicz spaces, see the monograph [43]. The Orlicz spaces are influential due to their several significant properties. For instance, the Hardy–Littlewood maximal operator is not bounded in $L^1(\mathbb{R}^n)$ but well-defined in Orlicz space $L^\phi(\mathbb{R}^n)$ for some convex function ϕ (see [23]). The theory of Orlicz spaces $L^\phi(\mathbb{R}^n)$ not only encompasses the classical L^p-spaces, but also incorporates other significant function spaces, including *Logarithmic space* and *Exponential space*. These function spaces play a crucial role in various branches of analysis. For instance, *Logarithmic space* provides a convenient frame to analyze the Hardy–Littlewood maximal function (see [16, 46]) and the *Exponential space* is important to the theory of Sobolev space embedding (see [14]). Hence the study on Orlicz space widens the range of applications of the presented theory, and this paper provides a unified approach to study the sampling problem in various function spaces simultaneously.

The sampling problem has not been explored much in the framework of Orlicz space. Recently, Pawlewicz and Wojciechowski [39] studied the sampling problem for the space of trigonometric polynomials and generalized the Marcinkiewicz sampling theorem for $L^\phi([0,2\pi])$. In this article, we generalize the sampling result of Nashed and Sun [33] for the image space V of an idempotent integral operator on the Orlicz space $L^\phi(\mathbb{R}^n)$ and prove the following inequality:

$$A\|f\|_{L^\phi(\mathbb{R}^n)} \leq \|(f(\gamma))_{\gamma\in\Gamma}\|_{\ell^\phi_w(\Gamma)} \leq B\|f\|_{L^\phi(\mathbb{R}^n)} \quad \forall f \in V, \tag{1.1}$$

where A, B are positive constants and $\ell^\phi_w(\Gamma)$ is a weighted Orlicz space with weight $w := (w_\gamma)_{\gamma\in\Gamma}$.

The signals are used to recover from their random measurements in the field of compressed sensing [15], learning theory [10, 11], and image processing [9]. The regularization approach in learning theory can be observed in [12, 20]. Some recent advancements in deep learning schemes are discussed in [30, 31, 40, 49].

In this paper, we further investigate the problem of sampling when the sample positions are selected randomly based on some probability distribution. This kind of sampling problem is commonly known as *random sampling*. It involves estimating the probability for which the sampling inequality (1.1) holds for V when the sample points are chosen randomly. Bass and Gröchenig [5] showed that the stable recovery of a band-limited signal from their random samples distributed over $\mathbb{R}^n$ is almost impossible. The positive result in case of random sampling was obtained when the random samples were distributed on a bounded subset $C_R := [-\frac{R}{2}, \frac{R}{2}]^n \subset \mathbb{R}^n$, and the sampling inequality (1.1) held for the class of functions concentrated on C_R, see [5, 6]. The random sampling problem was extensively studied for the reproducing kernel subspace of L^p-space, such as multivariate trigonometric space [4, 27], multiply generated shift-invariant spaces [17, 48], and image space of an idempotent integral operator [18, 28, 36, 38], and localized reproducing kernel space [37].

The paper is organized as follows. We begin with some preliminaries on Orlicz space, basic definitions and auxiliary results in Sec. 2. The notion of shift-invariant spaces in the Orlicz space has been discussed in Sec. 3. In Sec. 4, we first discuss the boundedness of an idempotent integral operator on T in $L^\phi(\mathbb{R}^n)$. Further, we derive the sampling inequality (1.1) for reproducing kernel subspace V of $L^\phi(\mathbb{R}^n)$ considered as the image of an idempotent integral operator. In Sec. 5, we prove that the sampling inequality (1.1) holds with overwhelming probability for the class of concentrated functions on C_R.

2. Preliminaries

Definition 2.1. A non-decreasing convex function $\phi : [0,\infty) \to [0,\infty]$ is said to be *Orlicz function* if

(a) $\phi(0) = 0$ and $\lim_{x\to\infty} \phi(x) = \infty$,
(b) ϕ is left continuous.

A few examples of the Orlicz functions and the corresponding Orlicz spaces $L^\phi(\mathbb{R}^n)$ are listed as follows.

(1) $\phi(x) = x^p/p$ with $1 \le p < \infty$; $L^\phi(\mathbb{R}^n) = L^p(\mathbb{R}^n)$.
(2) $\phi(x) = (x-1)\chi_{[1,\infty)}(x)$; $L^\phi(\mathbb{R}^n) = L^1(\mathbb{R}^n) + L^\infty(\mathbb{R}^n)$.
(3) $\phi(x) = x\log x$; $L^\phi(\mathbb{R}^n) =$ Logarithmic space $L\log L(\mathbb{R}^n)$.
(4) $\phi(x) = \exp(x^a) - 1$ with $a > 0$; $L^\phi(\mathbb{R}^n) =$ Exponential space $\exp L^a$.

In particular, for $n > 1$ and open and bounded subset Ω of $\mathbb{R}^n$ with Lipschitz boundary $\partial\Omega$, the Sobolev space $W_0^{1,n}(\Omega)$ is embedded in Orlicz space $L^\phi(\Omega)$ for $\phi(x) = \exp(x^{n/(n-1)}) - 1$.

Definition 2.2. The function ψ is said to be *Orlicz conjugate* of an Orlicz function ϕ if

$$\psi(x) := \sup_{0\le y<\infty} (xy - \phi(y)).$$

In the following, we mention some Orlicz functions and their Orlicz conjugate.

(1) For $\phi(x) = x^p/p$ with $1 < p < \infty$, the Orlicz conjugate function $\psi(x) = x^q/q$, where $q = \frac{p}{p-1}$.
(2) For $\phi(x) = e^x - x - 1$, the Orlicz conjugate function $\psi(x) = (1+x)\ln(1+x) - x$.
(3) For $\phi(x) = x\ln(1+x)$, the Orlicz conjugate function $\psi(x) \approx \cosh x - 1$.

Let ψ be the Orlicz conjugate of Orlicz function ϕ. Then the *Hölder's inequality* in context of Orlicz space is given by

$$\|fg\|_{L^1(\mathbb{R}^n)} \le 2\|f\|_{L^\phi(\mathbb{R}^n)}\|g\|_{L^\psi(\mathbb{R}^n)},$$

where $f \in L^\phi(\mathbb{R}^n)$ and $g \in L^\psi(\mathbb{R}^n)$.

Definition 2.3. We say $\phi \in \Delta_2$ if ϕ is an Orlicz function that satisfies

$$\phi(2x) \le L\,\phi(x), \quad \forall\, x > 0,$$

for some $L > 0$.

For example, $\phi(x) = x^p, 1 \le p < \infty$, and $\phi(x) = x(\log(e+x))$ are in Δ_2, but $\phi(x) = e^x - x - 1$ does not in Δ_2.

It is important to note that if $\phi \in \Delta_2$ then the dual space of $L^\phi(\mathbb{R}^n)$ is identical with $L^\psi(\mathbb{R}^n)$, where ψ is the Orlicz conjugate of ϕ. In addition, if $\psi \in \Delta_2$ then $L^\phi(\mathbb{R}^n)$ is a reflexive Banach space (see [41]). Moreover, in terms of the norm topology given by the norm $\|\cdot\|_{L^\phi(\mathbb{R}^n)}$, the set $B_\phi = \{f \in L^\phi(\mathbb{R}^n) : \|f\|_{L^\phi(\mathbb{R}^n)} < 1\}$ is an open set if and only if $\phi \in \Delta_2$.

The Orlicz sequence space corresponding to the discrete set Γ is defined by

$$\ell^\phi(\Gamma) := \left\{(c_\gamma)_{\gamma\in\Gamma} : \sum_{\gamma\in\Gamma} \phi\left(\frac{|c_\gamma|}{\lambda}\right) < \infty \text{ for some } \lambda > 0\right\},$$

with the norm

$$\|(c_\gamma)_{\gamma\in\Gamma}\|_{\ell^\phi(\Gamma)} := \inf\left\{\lambda > 0 : \sum_{\gamma\in\Gamma} \phi\left(\frac{|c_\gamma|}{\lambda}\right) \le 1\right\}.$$

Let $w = (w_\gamma)_{\gamma\in\Gamma}$ be a weight sequence. Then, the corresponding weighted Orlicz sequence space is defined by

$$\ell_w^\phi(\Gamma) := \left\{(c_\gamma)_{\gamma\in\Gamma} : \sum_{\gamma\in\Gamma} \phi\left(\frac{|c_\gamma|}{\lambda}\right) w_\gamma < \infty \text{ for some } \lambda > 0\right\},$$

with the norm

$$\|(c_\gamma)_{\gamma\in\Gamma}\|_{\ell_w^\phi(\Gamma)} := \inf\left\{\lambda > 0 : \sum_{\gamma\in\Gamma} \phi\left(\frac{|c_\gamma|}{\lambda}\right) w_\gamma \le 1\right\}.$$

For any $a := (a_m)_{m\in\mathbb{Z}^n} \in \ell^\phi(\mathbb{Z}^n)$ and $b := (b_m)_{m\in\mathbb{Z}^n} \in \ell^1(\mathbb{Z}^n)$, the convolution sequence of a and b is denoted by $a * b$, and the mth term of $a * b$ is defined by

$$(a * b)_m = \sum_{i\in\mathbb{Z}^n} a_{m-i} b_i.$$

Lemma 2.1. *Let ϕ be an Orlicz function. For every $a := (a_m)_{m\in\mathbb{Z}^n} \in \ell^\phi(\mathbb{Z}^n)$ and $b := (b_m)_{m\in\mathbb{Z}^n} \in \ell^1(\mathbb{Z}^n)$, the convolution sequence $a * b \in \ell^\phi(\mathbb{Z}^n)$ and*

$$\|a * b\|_{\ell^\phi(\mathbb{Z}^n)} \le \|a\|_{\ell^\phi(\mathbb{Z}^n)} \|b\|_{\ell^1(\mathbb{Z}^n)}.$$

Proof. Let $a := (a_m)_{m\in\mathbb{Z}^n} \in \ell^\phi(\mathbb{Z}^n)$ and $b := (b_m)_{m\in\mathbb{Z}^n} \in \ell^1(\mathbb{Z}^n)$. Further, we assume that $\lambda = \|a\|_{\ell^\phi(\mathbb{Z}^n)}\|b\|_{\ell^1(\mathbb{Z}^n)}$. In view of Jensen's inequality and Fubini's theorem, we get

$$\begin{aligned}
\sum_{m\in\mathbb{Z}^n} \phi\left(\frac{|(a*b)_m|}{\lambda}\right) &= \sum_{m\in\mathbb{Z}^n} \phi\left(\frac{\sum_{i\in\mathbb{Z}^n} |a_{m-i} b_i|}{\|a\|_{\ell^\phi(\mathbb{Z}^n)}\|b\|_{\ell^1(\mathbb{Z}^n)}}\right) \\
&\le \sum_{m\in\mathbb{Z}^n} \sum_{i\in\mathbb{Z}^n} \frac{|b_i|}{\|b\|_{\ell^1(\mathbb{Z}^n)}} \phi\left(\frac{|a_{m-i}|}{\|a\|_{\ell^\phi(\mathbb{Z}^n)}}\right) \\
&\le \sum_{i\in\mathbb{Z}^n} \frac{|b_i|}{\|b\|_{\ell^1(\mathbb{Z}^n)}} \sum_{m\in\mathbb{Z}^n} \phi\left(\frac{|a_{m-i}|}{\|a\|_{\ell^\phi(\mathbb{Z}^n)}}\right) \le 1.
\end{aligned}$$

This gives $\|a * b\|_{\ell^\phi(\mathbb{Z}^n)} \le \|a\|_{\ell^\phi(\mathbb{Z}^n)}\|b\|_{\ell^1(\mathbb{Z}^n)}$. $\square$

In the following, we recall some preliminary definitions that will be used in the paper.

Definition 2.4. A countable set $\Gamma \subset \mathbb{R}^n$ is said to be *relatively separated* with gap $\eta > 0$ if

$$A_\Gamma(\eta) := \inf_{x\in\mathbb{R}^n} \sum_{\gamma\in\Gamma} \chi_{B(\gamma;\frac{\eta}{2})}(x) \geq 1, \tag{2.1}$$

$$B_\Gamma(\eta) := \sup_{x\in\mathbb{R}^n} \sum_{\gamma\in\Gamma} \chi_{B(\gamma;\frac{\eta}{2})}(x) < \infty, \tag{2.2}$$

where $B(\gamma; r)$ denotes an open cube of side-length $2r$ with respect to $\|\cdot\|_\infty$ in $\mathbb{R}^n$.

Definition 2.5. Let (X, d) be a metric space and $\epsilon > 0$. A subset $S \subset X$ is said be ϵ-net of X if $X \subseteq \bigcup_{x\in S} B(x;\epsilon)$. Moreover, S is said to be a finite ϵ-net of X if S is finite.

Definition 2.6. Let Γ be a relatively separated subset of $\Omega \subseteq \mathbb{R}^n$ with gap η. A Bounded Uniform Partition of Unity (BUPU) of Ω associated with the covering $\{B(\gamma;\eta/2) : \gamma \in \Gamma\}$ is the collection of functions $\{\beta_\gamma\}_{\gamma\in\Gamma}$ such that

(1) $\sum_{\gamma\in\Gamma} \beta_\gamma(x) = 1, \forall\, x \in \Omega,$
(2) $0 \leq \beta_\gamma(x) \leq 1, \forall\, x \in \Omega, \forall\, \gamma \in \Gamma,$
(3) for each $\gamma \in \Gamma,$ supp $(\beta_\gamma) \subseteq B(\gamma;\eta/2)$.

Here, $B(x;\epsilon)$ denotes the open ball of radius ϵ centered at x in $\mathbb{R}^n$.

3. Shift Invariant Spaces in the Orlicz Space

In this section, we discuss the notion of shift-invariant space in the Orlicz space setting. The shift-invariant subspace of $L^p(\mathbb{R}^n)$ can be written as

$$V_p(h) := \left\{ \sum_{i\in\mathbb{Z}^n} c_i h(\cdot - i) : (c_i)_{i\in\mathbb{Z}^n} \in \ell^p(\mathbb{Z}^n) \right\},$$

under certain conditions on the generator h. To the best of our knowledge, the notion of shift-invariant space in the Orlicz space $L^\phi(\mathbb{R}^n)$ has not been studied in the literature. In this section, we discuss the condition on the generator h for which the integer shifts of the generator form a Riesz-type sequence for the shift-invariant space in $L^\phi(\mathbb{R}^n)$.

Definition 3.1. The Wiener amalgam space $W(L^\infty, \ell^p)(\mathbb{R}^n)$ contains all measurable functions such that

$$\|f\|^p_{W(L^\infty,\ell^p)(\mathbb{R}^n)} := \sum_{i\in\mathbb{Z}^n} \sup_{x\in[0,1]^n} |f(x+i)|^p < \infty.$$

Proposition 3.1 ([21, 26]). *Assume that $h \in W(L^\infty, \ell^1)(\mathbb{R}^n)$ satisfies*

$$m\|c\|_{\ell^2(\mathbb{Z}^n)} \le \left\| \sum_{i\in\mathbb{Z}^n} c_i h(\cdot - i) \right\|_{L^2(\mathbb{R}^n)} \le M\|c\|_{\ell^2(\mathbb{Z}^n)}, \quad \forall\, c = (c_i)_{i\in\mathbb{Z}^n} \in \ell^2(\mathbb{Z}^n), \tag{3.1}$$

where $0 < m \le M$. Then there exists a function $\tilde{h} \in W(L^\infty, \ell^1)(\mathbb{R}^n)$ such that

$$\int_{\mathbb{R}^n} h(y-i)\tilde{h}(y-j)dy = \delta_{i,j},$$

where $\delta_{i,j}$ is the Kronecker delta function.

Theorem 3.1. *Under the assumptions of Proposition* 3.1, *the following inequality holds*

$$C_1\|c\|_{\ell^\phi(\mathbb{Z}^n)} \le \left\| \sum_{i\in\mathbb{Z}^n} c_i h(\cdot - i) \right\|_{L^\phi(\mathbb{R}^n)} \le C_2\|c\|_{\ell^\phi(\mathbb{Z}^n)}, \quad \forall\, c = (c_i)_{i\in\mathbb{Z}^n} \in \ell^\phi(\mathbb{Z}^n),$$

where $C_1, C_2 > 0$ and ϕ is an Orlicz function.

Proof. Let λ be a positive real. Then, we can write

$$\begin{aligned}
\int_{\mathbb{R}^n} \phi\left(\frac{\left|\sum_{i\in\mathbb{Z}^n} c_i h(x-i)\right|}{\lambda} \right) dx &= \sum_{j\in\mathbb{Z}^n} \int_{[0,1]^n} \phi\left(\frac{\left|\sum_{i\in\mathbb{Z}^n} c_i h(x+i-j)\right|}{\lambda} \right) dx \\
&\le \sum_{j\in\mathbb{Z}^n} \phi\left(\frac{\sum_{i\in\mathbb{Z}^n} |c_i| \sup_{x\in[0,1]^n} |h(x+i-j)|}{\lambda} \right) \\
&\le \sum_{j\in\mathbb{Z}^n} \phi\left(\frac{|c| * \sup_{x\in[0,1]^n} |h(x+\cdot)|}{\lambda} \right).
\end{aligned}$$

From Lemma 2.1, we have

$$\begin{aligned}
\left\| \sum_{i\in\mathbb{Z}^n} c_i h(\cdot - i) \right\|_{L^\phi(\mathbb{R}^n)} &\le \left\| |c| * \sup_{x\in[0,1]^n} |h(x+\cdot)| \right\|_{\ell^\phi(\mathbb{Z}^n)} \\
&\le \|c\|_{\ell^\phi(\mathbb{Z}^n)} \left\| \sup_{x\in[0,1]^n} |h(x+\cdot)| \right\|_{\ell^1(\mathbb{Z}^n)} \\
&\le \|c\|_{\ell^\phi(\mathbb{Z}^n)} \|h\|_{W(L^\infty, \ell^1)(\mathbb{R}^n)}.
\end{aligned} \tag{3.2}$$

In order to prove $\|c\|_{\ell^\phi(\mathbb{Z}^n)} \lesssim \|f\|_{L^\phi(\mathbb{R}^n)}$ for any $c = (c_i)_{i\in\mathbb{Z}^n} \in \ell^\phi(\mathbb{Z}^n)$ and $f = \sum_{i\in\mathbb{Z}^n} c_i h(\cdot\ - i)$, we use the equivalent definition of $\|\cdot\|_{\ell^\phi(\mathbb{Z}^n)}$, i.e.

$$\|c\|_{\ell^\phi(\mathbb{R}^n)} = \sup\left\{\left|\sum_{i\in\mathbb{Z}^n} c_i d_i\right| : \|(d_i)_{i\in\mathbb{Z}^n}\|_{\ell^\psi(\mathbb{Z}^n)} \le 1\right\},$$

where ψ is the Orlicz conjugate of ϕ. By Proposition 3.1, there exists a dual $\tilde{h} \in W(L^\infty, \ell^1)(\mathbb{R}^n)$ such that

$$\int_{\mathbb{R}^n} h(x-i)\tilde{h}(x)dx = \delta_{i,0}.$$

This implies $c_i = \int_{\mathbb{R}^n} f(x)\tilde{h}(x-i)dx$. Now assume that $d = (d_i)_{i\in\mathbb{Z}^n} \in \ell^\psi(\mathbb{Z}^n)$. Then we have

$$\begin{aligned}
\left|\sum_{i\in\mathbb{Z}^n} c_i d_i\right| &= \left|\sum_{i\in\mathbb{Z}^n} d_i \int_{\mathbb{R}^n} f(x)\tilde{h}(x-i)dx\right| \\
&= \left|\int_{\mathbb{R}^n} f(x) \sum_{i\in\mathbb{Z}^n} d_i\tilde{h}(x-i)dx\right| \\
&\le 2\|f\|_{L^\phi(\mathbb{R}^n)} \left\|\sum_{i\in\mathbb{Z}^n} d_i\tilde{h}(\cdot - i)\right\|_{L^\psi(\mathbb{R}^n)} \\
&\le 2\|d\|_{\ell^\psi(\mathbb{Z}^n)}\|\tilde{h}\|_{W(L^\infty,\ell^1)(\mathbb{R}^n)} \left\|\sum_{i\in\mathbb{Z}^n} c_i h(\cdot - i)\right\|_{L^\phi(\mathbb{R}^n)} \qquad \text{(by (3.2))}.
\end{aligned}$$

Therefore, we obtain

$$\|c\|_{\ell^\phi(\mathbb{Z}^n)} \le 2\|\tilde{h}\|_{W(L^\infty,\ell^1)(\mathbb{R}^n)} \left\|\sum_{i\in\mathbb{Z}^n} c_i h(\cdot - i)\right\|_{L^\phi(\mathbb{R}^n)}.$$

This establishes the result. □

Remark 3.1. Let ϕ be an Orlicz function and $h \in W(L^\infty, \ell^1)(\mathbb{R}^n)$ that satisfies (3.1) then the shift-invariant space

$$V_\phi(h) := \left\{\sum_{i\in\mathbb{Z}^n} c_i\ h(\cdot - i) : (c_i)_{i\in\mathbb{Z}^n} \in \ell^\phi(\mathbb{Z}^n)\right\},$$

is a closed subspace of $L^\phi(\mathbb{R}^n)$. Moreover, there exists a dual $\tilde{h} \in W(L^\infty, \ell^1)(\mathbb{R}^n)$ such that for $x, y \in \mathbb{R}^n$,

$$K(x,y) = \sum_{i\in\mathbb{Z}^n} h(x-i)\tilde{h}(y-i).$$

Then $V_\phi(h)$ can be written as

$$V_\phi(h) = \{Tf : f \in L^\phi(\mathbb{R}^n)\},$$

where T is an idempotent integral operator with integral kernel K.

4. Sampling Inequality

In this section, we generalize the notion of shift-invariant space in a more general setting. We define the idempotent integral operator on Orlicz space $L^\phi(\mathbb{R}^n)$ and establish the sampling inequality for the image space of the integral operator.

Let $T : L^\phi(\mathbb{R}^n) \to L^\phi(\mathbb{R}^n)$ be an idempotent integral operator ($T^2 = T$) defined as

$$Tf(x) = \int_{\mathbb{R}^n} K(x,y)f(y)dy, \quad x \in \mathbb{R}^n, \tag{4.1}$$

where K is a continuous function. In addition, we assume that the integral kernel K satisfies off-diagonal decay condition

$$\|K\|_S := \max\left\{\sup_{x\in\mathbb{R}^n}\int_{\mathbb{R}^n}|K(x,y)|dy, \sup_{y\in\mathbb{R}^n}\int_{\mathbb{R}^n}|K(x,y)|dx\right\} < \infty \tag{4.2}$$

and the regularity condition

$$\lim_{\eta\to 0}\|\omega_\eta(K)\|_S = 0, \tag{4.3}$$

where for $x, y \in \mathbb{R}^n$

$$\omega_\eta(K)(x,y) := \sup_{\bar{x},\bar{y}\in[-\frac{\eta}{2},\frac{\eta}{2}]^n} |K(x+\bar{x}, y+\bar{y}) - K(x,y)|.$$

In order to show that the integral operator T is a bounded operator, we recall the following result.

Lemma 4.1 ([29]). *Let ϕ be an Orlicz function and $F : L^1(\mathbb{R}^n) + L^\infty(\mathbb{R}^n) \to L^1(\mathbb{R}^n) + L^\infty(\mathbb{R}^n)$ be a linear operator such that F maps $L^1(\mathbb{R}^n)$ into $L^1(\mathbb{R}^n)$ and $L^\infty(\mathbb{R}^n)$ into $L^\infty(\mathbb{R}^n)$ are bounded operator with*

$$\|Ff\|_{L^1(\mathbb{R}^n)} \le M\|f\|_{L^1(\mathbb{R}^n)} \quad \forall f \in L^1(\mathbb{R}^n),$$
$$\|Ff\|_{L^\infty(\mathbb{R}^n)} \le M\|f\|_{L^\infty(\mathbb{R}^n)} \quad \forall f \in L^\infty(\mathbb{R}^n).$$

Then for all $f \in L^1(\mathbb{R}^n) \cap L^\phi(\mathbb{R}^n)$,

$$\|Ff\|_{L^\phi(\mathbb{R}^n)} \le M\|f\|_{L^\phi(\mathbb{R}^n)}.$$

Theorem 4.1. *Let $\phi \in \Delta_2$ and T be an idempotent integral operator on $L^\phi(\mathbb{R}^n)$ as defined in* (4.1) *with the kernel K that satisfies* (4.2)*. Then T is a bounded operator,* *and*

$$\|Tf\|_{L^\phi(\mathbb{R}^n)} \leq \|K\|_S \|f\|_{L^\phi(\mathbb{R}^n)} \quad \forall f \in L^\phi(\mathbb{R}^n). \tag{4.4}$$

Proof. For $1 \leq p \leq \infty$ and $f \in L^p(\mathbb{R}^n)$, the following estimate holds (see [28])

$$\|Tf\|_{L^p(\mathbb{R}^n)} \leq \|K\|_S \|f\|_{L^p(\mathbb{R}^n)}.$$

Let $\phi \in \Delta_2$. Then $L^1(\mathbb{R}^n) \cap L^\phi(\mathbb{R}^n)$ is a dense subset of $L^\phi(\mathbb{R}^n)$ (see [43]). Hence, the bounded operator $T : L^1(\mathbb{R}^n) \cap L^\phi(\mathbb{R}^n) \to L^\phi(\mathbb{R}^n)$ obtained in Proposition 4.1 can be extended continuously to $T : L^\phi(\mathbb{R}^n) \to L^\phi(\mathbb{R}^n)$. □

For a given sample set $\Gamma \subset \mathbb{R}^n$, every function f in a given function space is measured at $\gamma \in \Gamma$. Hence the point evaluation functionals on a closed space $W \subseteq L^\phi(\mathbb{R}^n)$ must be continuous, i.e. for each $x \in \mathbb{R}^n$, there exists $M_x > 0$ such that

$$|f(x)| \leq M_x \|f\|_{L^\phi(\mathbb{R}^n)}, \quad \forall f \in W.$$

The closed subspace W is called *reproducing kernel subspace* of $L^\phi(\mathbb{R}^n)$.

We denote V as the image space of idempotent integral operator T on $L^\phi(\mathbb{R}^n)$, i.e.

$$V := \{f \in L^\phi(\mathbb{R}^n) : Tf = f\}.$$

Theorem 4.2. *Under the assumptions of Theorem* 4.1, V *is a reproducing kernel subspace of* $L^\phi(\mathbb{R}^n)$.

Proof. Let $x \in \mathbb{R}^n$ and $\phi \in \Delta_2$. Then by using Hölder's inequality, we obtain

$$|f(x)| \leq 2\|K(x,\cdot)\|_{L^\psi(\mathbb{R}^n)} \|f\|_{L^\phi(\mathbb{R}^n)}, \quad \forall f \in V,$$

where ψ is Orlicz conjugate of ϕ. If X is a symmetric space, then from the embedding theorem [43, Theorem 5.2.1], we have

$$L^1(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n) \subseteq X$$

and

$$\|g\|_X \leq 2\max\{\|g\|_{L^1(\mathbb{R}^n)}, \|g\|_{L^\infty(\mathbb{R}^n)}\}, \quad \forall g \in L^1(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n).$$

As the kernel K satisfies (4.2), K is bounded (see [28]). In particular, there exists $C > 1$ such that

$$\|K(x,.)\|_{L^1(\mathbb{R}^n)} \leq \|K\|_S, \quad \text{and} \quad \|K(x,.)\|_{L^\infty(\mathbb{R}^n)} \leq C\|K\|_S.$$

Since $L^\psi(\mathbb{R}^n)$ is a symmetric space, we have $K(x,\cdot) \in L^\psi(\mathbb{R}^n)$, and $\|K(x,\cdot)\|_{L^\psi(\mathbb{R}^n)} \leq 2C\|K\|_S$. This implies

$$|f(x)| \leq 4C\|K\|_S \|f\|_{L^\phi(\mathbb{R}^n)} \quad \forall f \in V.$$

This completes the proof. □

Example 4.1. Let Γ be a relatively separated subset of $\mathbb{R}^n$, and ψ be the Orlicz conjugate of Orlicz function ϕ. Assume that the two families $H = \{h_\gamma\}_{\gamma\in\Gamma} \subset L^\phi(\mathbb{R}^n)$ and $\widetilde{H} = \{\tilde{h}_\gamma\}_{\gamma\in\Gamma} \subset L^\psi(\mathbb{R}^n)$ satisfy

$$\int_{\mathbb{R}^n} h_\gamma(x)\tilde{h}_{\gamma'}(x)dx = \delta_{\gamma,\gamma'},$$

for each $\gamma, \gamma' \in \Gamma$. Then, the integral operator associated with the kernel

$$K(x,y) = \sum_{\gamma\in\Gamma} h_\gamma(x)\tilde{h}_\gamma(y)$$

is idempotent. Moreover, if H and $\widetilde{H}$ are localized, i.e. there exists $f \in W(L^\infty, \ell^1)(\mathbb{R}^n)$ such that

$$|h_\gamma(x)| + |\tilde{h}_\gamma(x)| \leq f(x-\gamma), \quad x \in \mathbb{R}^n, \ \gamma \in \Gamma,$$

and for $\epsilon \in (0,1)$ there exists $g \in W(L^\infty, \ell^1)(\mathbb{R}^n)$ with $\|g\|_{W(L^\infty,\ell^1)(\mathbb{R}^n)} < \epsilon$ such that

$$|\omega_\epsilon(h_\gamma)(x)| + |\omega_\epsilon(\tilde{h}_\gamma)(x)| \leq g(x-\gamma), \quad x \in \mathbb{R}^n, \ \gamma \in \Gamma,$$

then the kernel satisfies (4.2) and (4.3).

Example 4.2. Let σ_n be the surface area of the unit sphere $\mathbb{S}^{n-1} \subset \mathbb{R}^n$ and $\|\cdot\|_2$ be the square norm on $\mathbb{R}^n$. Consider

$$K(x,y) = \frac{2}{\sigma_n}\sqrt{\frac{2}{\pi}}\exp(-\|x\|_2^2 - \|y\|_2^2),$$

then the integral operator T defined in (4.1), is an idempotent operator. In fact, T is a bounded operator on $L^\phi(\mathbb{R}^n)$ as $\|K\|_S = \sqrt{2}$. Moreover, one can verify that for a small value of $\eta > 0$,

$$\|\omega_\eta(K)\|_S \leq C\eta,$$

where C is a positive constant.

In the following theorem, we show the existence of a stable sampling set for the space V.

Theorem 4.3. *Let $\phi \in \Delta_2$ and K be the integral kernel of the idempotent operator T defined in* (4.1) *that satisfies* (4.2) *and* (4.3). *Further we assume that there exists $\eta > 0$ such that*

$$\|\omega_\eta(K)\|_S < 1. \tag{4.5}$$

Then, any signal $f \in V$ can be reconstructed stably from its discrete measurement on a relatively separated set Γ with gap η. Moreover, the sampling inequality

$$\begin{aligned}(1 - \|\omega_\eta(K)\|_S)\|f\|_{L^\phi(\mathbb{R}^n)} &\leq \|(f(\gamma))_{\gamma\in\Gamma}\|_{\ell^\phi_w(\Gamma)} \\ &\leq 2(1 + \|\omega_\eta(K)\|_S)\|f\|_{L^\phi(\mathbb{R}^n)} \end{aligned} \tag{4.6}$$

holds for all $f \in V$. Here the weight $w = (\|\beta_\gamma\|_{L^1(\mathbb{R}^n)})_{\gamma\in\Gamma}$ and $\{\beta_\gamma\}_{\gamma\in\Gamma}$ is a BUPU of $\mathbb{R}^n$ associated with the covering $\{B(\gamma;\eta/2) : \gamma \in \Gamma\}$.

Proof. Let $f \in V$ and $\Gamma \subseteq \mathbb{R}^n$ be a relatively separated set with gap η that satisfies (4.5). We define modulus of continuity of f as follows:

$$\omega_\eta(f)(x) := \sup_{\bar{x}\in[-\frac{\eta}{2},\frac{\eta}{2}]^n} |f(x+\bar{x}) - f(x)|.$$

For $\gamma \in \Gamma$ and $x \in B(\gamma;\eta/2)$, we can write

$$|f(x)| - \omega_\eta(f)(x) \le |f(\gamma)| \le |f(x)| + \omega_\eta(f)(x).$$

As $\{\beta_\gamma\}_{\gamma\in\Gamma}$ is a BUPU of $\mathbb{R}^n$ associated with the covering $\{B(\gamma;\eta/2) : \gamma \in \Gamma\}$, we get

$$\begin{aligned}
\|f\|_{L^\phi(\mathbb{R}^n)} &\le \left\|\sum_{\gamma\in\Gamma}(|f(\gamma)| + \omega_\eta(f))\beta_\gamma\right\|_{L^\phi(\mathbb{R}^n)} \\
&\le \left\|\sum_{\gamma\in\Gamma}|f(\gamma)|\beta_\gamma\right\|_{L^\phi(\mathbb{R}^n)} + \left\|\sum_{\gamma\in\Gamma}\omega_\eta(f)\beta_\gamma\right\|_{L^\phi(\mathbb{R}^n)} \\
&\le \|(f(\gamma))_{\gamma\in\Gamma}\|_{\ell^\phi_w(\Gamma)} + \|\omega_\eta(f)\|_{L^\phi(\mathbb{R}^n)},
\end{aligned}$$

as $\int_{\mathbb{R}^n} \phi(\frac{\sum_{\gamma\in\Gamma}|f(\gamma)|\beta_\gamma(x)}{\lambda})dx \le \sum_{\gamma\in\Gamma}\phi(\frac{|f(\gamma)|}{\lambda})\|\beta_\gamma\|_{L^1(\mathbb{R}^n)}$.

This gives

$$\|(f(\gamma))_{\gamma\in\Gamma}\|_{\ell^\phi_w(\Gamma)} \ge \|f\|_{L^\phi(\mathbb{R}^n)} - \|\omega_\eta(f)\|_{L^\phi(\mathbb{R}^n)},$$

where the weight sequence $w = (\|\beta_\gamma\|_{L^1(\mathbb{R}^n)})_{\gamma\in\Gamma}$.

To prove right-hand side of the sampling inequality,

$$\begin{aligned}
\sum_{\gamma\in\Gamma}\phi\left(\frac{|f(\gamma)|}{\lambda}\right)\|\beta_\gamma\|_{L^1(\mathbb{R}^n)} &= \sum_{\gamma\in\Gamma}\phi\left(\frac{|f(\gamma)|}{\lambda}\right)\int_{\mathbb{R}^n}\beta_\gamma(x)dx \\
&\le \sum_{\gamma\in\Gamma}\int_{[\gamma-\frac{\eta}{2},\gamma+\frac{\eta}{2}]^n}\phi\left(\frac{|f(\gamma)|}{\lambda}\right)\beta_\gamma(x)dx \\
&\le \sum_{\gamma\in\Gamma}\int_{[\gamma-\frac{\eta}{2},\gamma+\frac{\eta}{2}]^n}\phi\left(\frac{|f(x)| + \omega_\eta(f)(x)}{\lambda}\right)\beta_\gamma(x)dx \\
&\le \int_{\mathbb{R}^n}\phi\left(\frac{|f(x)| + \omega_\eta(f)(x)}{\lambda}\right)dx.
\end{aligned}$$

The convexity of ϕ implies

$$\sum_{\gamma\in\Gamma}\phi\left(\frac{|f(\gamma)|}{\lambda}\right)\|\beta_\gamma\|_{L^1(\mathbb{R}^n)} \le \frac{1}{2}\int_{\mathbb{R}^n}\phi\left(\frac{2|f(x)|}{\lambda}\right)dx + \frac{1}{2}\int_{\mathbb{R}^n}\phi\left(\frac{2\omega_\eta(f)(x)}{\lambda}\right)dx.$$

Assume that

$$\lambda_1 = \inf\left\{\tilde{\lambda}>0 : \int_{\mathbb{R}^n}\phi\left(\frac{2|f(x)|}{\tilde{\lambda}}\right)dx \le 1\right\},$$
$$\lambda_2 = \inf\left\{\tilde{\lambda}>0 : \int_{\mathbb{R}^n}\phi\left(\frac{2\omega_\eta(f)(x)}{\tilde{\lambda}}\right)dx \le 1\right\}$$

and consider $\lambda = \lambda_1 + \lambda_2$. Using monotonicity of the convex function ϕ, we have

$$\begin{aligned}\sum_{\gamma\in\Gamma}\phi\left(\frac{|f(\gamma)|}{\lambda}\right)\|\beta_\gamma\|_{L^1(\mathbb{R}^n)} &\le \frac{1}{2}\int_{\mathbb{R}^n}\phi\left(\frac{2|f(x)|}{\lambda_1}\right)dx + \frac{1}{2}\int_{\mathbb{R}^n}\phi\left(\frac{2\omega_\eta(f)(x)}{\lambda_2}\right)dx \\ &\le 1.\end{aligned}$$

This gives

$$\|(f(\gamma))_{\gamma\in\Gamma}\|_{\ell^\phi_w(\Gamma)} \le \|2f\|_{L^\phi(\mathbb{R}^n)} + \|2\omega_\eta(f)\|_{L^\phi(\mathbb{R}^n)}.$$

It can be easily verified that for every $x\in\mathbb{R}^n$, $\omega_\eta(f)(x) \le \int_{\mathbb{R}^n}\omega_\eta(K)(x,y)|f(y)|dy$. Therefore, (4.4) implies

$$\|\omega_\eta(f)\|_{L^\phi(\mathbb{R}^n)} \le \|\omega_\eta(K)\|_S\|f\|_{L^\phi(\mathbb{R}^n)}.$$

Hence, we deduce the following inequality

$$(1-\|\omega_\eta(K)\|_S)\|f\|_{L^\phi(\mathbb{R}^n)} \le \|(f(\gamma))_{\gamma\in\Gamma}\|_{\ell^\phi_w(\Gamma)} \le 2(1+\|\omega_\eta(K)\|_S)\|f\|_{L^\phi(\mathbb{R}^n)}.$$

This completes the proof. □

Remark 4.1. For $0<\eta<1$ that satisfies (4.5) and any relatively separated set Γ with gap η, we define a BUPU (see Definition 2.6) of $\mathbb{R}^n$ associated with the covering $\{B(\gamma;\eta/2):\gamma\in\Gamma\}$ as follows:

$$\beta_\gamma(x) := \frac{\chi_{B(\gamma;\eta/2)}(x)}{\sum_{\gamma'\in\Gamma}\chi_{B(\gamma';\eta/2)}(x)}, \quad \gamma\in\Gamma.$$

Then we have

$$\frac{\eta^n}{B_\Gamma(\eta)} \le \|\beta_\gamma\|_{L^1(\mathbb{R}^n)} \le \frac{\eta^n}{A_\Gamma(\eta)} \quad \forall\,\gamma\in\Gamma,$$

where $A_\Gamma(\eta)$ and $B_\Gamma(\eta)$ are as defined in (2.1) and (2.2), respectively. The sampling inequality (4.6) reduces to

$$\begin{aligned}(1-\|\omega_\eta(K)\|_S)\frac{A_\Gamma(\eta)}{\eta^n}\|f\|_{L^\phi(\mathbb{R}^n)} &\le \|(f(\gamma))_{\gamma\in\Gamma}\|_{\ell^\phi(\Gamma)}\\ &\le 2(1+\|\omega_\eta(K)\|_S)\frac{B_\Gamma(\eta)}{\eta^n}\|f\|_{L^\phi(\mathbb{R}^n)}.\end{aligned}$$

5. Random Sampling

In this section, we deal with the problem of random sampling for the set of functions that are δ-concentrated on C_R, i.e.

$$V(R,\delta) := \{f\in V : \|f\|_{L^\phi(C_R)} \ge (1-\delta)\|f\|_{L^\phi(\mathbb{R}^n)}\}.$$

We prove that the set of random samples X satisfies the sampling inequality (1.1) for $V(R,\delta)$ with an overwhelming probability, provided the sample size is at least of order $\mathcal{O}(R^n\ln(R^n))$. In addition, we assume that the regularity assumption (4.3) of the kernel K converges to zero linearly. In particular, we assume that there exists $\theta\in(0,1]$ such that

$$\kappa_\theta := \|K\|_S + \sup_{0<\eta\le 1}\eta^{-\theta}\|\omega_\eta(K)\|_S < \infty. \tag{5.1}$$

Definition 5.1. Let $X=\{x_j : 1\le j\le m\}$ be a finite subset of C_R. Then the *covering radius* for C_R associate with X is a positive constant denoted by $r_X(m)$, and it is defined as

$$r_X(m) := \inf\left\{r>0 : C_R\subseteq\bigcup_{j=1}^m B(x_j;r)\right\}.$$

The following theorem shows that under a sufficiently small covering radius of the sample set X on C_R, the sampling inequality holds for the set of δ-concentrated functions.

Theorem 5.1. *Let $\phi\in\Delta_2$, $\delta\in(0,1)$, and $X=\{x_j\in C_R : 1\le j\le m\}$ be a finite collection of sample points. Further we assume that T is the idempotent integral operator as defined in* (4.1) *with the kernel K that satisfies* (5.1), *and the covering radius for C_R associate with X satisfies*

$$r := r_X(m) < \left(\frac{1-\delta}{\kappa_\theta}\right)^{\frac{1}{\theta}}. \tag{5.2}$$

Then for all $f\in V(R,\delta)$, the following inequality holds:

$$(1-\delta-r^\theta\kappa_\theta)\|f\|_{L^\phi(\mathbb{R}^n)} \le \|(f(x_j))_{x_j\in X}\|_{\ell^\phi_w(X)} \le 2(1+r^\theta\kappa_\theta)\|f\|_{L^\phi(\mathbb{R}^n)}, \tag{5.3}$$

where the weight $w=(\|\beta_j\|_{L^1(\mathbb{R}^n)})_{j=1}^m$ and $\{\beta_j : 1\le j\le m\}$ is a BUPU of C_R associated with the covering $\{B(x_j;r) : 1\le j\le m\}$.

Proof. Let $X = \{x_j \in C_R : 1 \le j \le m\}$ be a finite collection of sample points, and r be the covering radius for C_R associated with X that satisfies (5.2). Then there exists a BUPU $\{\beta_j : 1 \le j \le m\}$ of C_R associate with the covering $\{B(x_j; r) : 1 \le j \le m\}$. We define an operator on V as follows:

$$Af := \sum_{j=1}^{m} f(x_j)\beta_j.$$

In view of Jensen's inequality, we have

$$\begin{aligned}\int_{C_R} \phi\left(\frac{|Af(x)|}{\lambda}\right)dx &= \int_{C_R} \phi\left(\frac{\left|\sum_{j=1}^{m} f(x_j)\beta_j(x)\right|}{\lambda}\right)dx \\ &\le \sum_{j=1}^{m} \int_{C_R} \beta_j(x)\phi\left(\frac{|f(x_j)|}{\lambda}\right)dx \\ &\le \sum_{j=1}^{m} \phi\left(\frac{|f(x_j)|}{\lambda}\right)\|\beta_j\|_{L^1(\mathbb{R}^n)}.\end{aligned}$$

Therefore, for $w = (\|\beta_j\|_{L^1(\mathbb{R}^n)})_{j=1}^m$ and any $f \in V$, we have

$$\|Af\|_{L^\phi(C_R)} \le \|(f(x_j))_{x_j \in X}\|_{\ell^\phi_w(X)}. \tag{5.4}$$

For $1 \le j \le m$, $\mathrm{supp}(\beta_j) \subseteq B(x_j; r)$ and $f \in V$, we obtain

$$\begin{aligned}|Af(x) - f(x)| &= \left|\int_{\mathbb{R}^n} \sum_{j=1}^{m}(K(x_j, y) - K(x, y))\beta_j(x)f(y)dy\right| \\ &\le \int_{\mathbb{R}^n} \omega_r(K)(x, y)|f(y)|dy.\end{aligned}$$

This implies

$$\|Af - f\|_{L^\phi(C_R)} \le r^\theta \kappa_\theta \|f\|_{L^\phi(\mathbb{R}^n)}.$$

In particular, for $f \in V(R, \delta)$, we have

$$\begin{aligned}\|Af\|_{L^\phi(C_R)} &\ge \|f\|_{L^\phi(C_R)} - \|f - Af\|_{L^\phi(C_R)} \\ &\ge (1 - \delta - r^\theta \kappa_\theta)\|f\|_{L^\phi(\mathbb{R}^n)}.\end{aligned}$$

Now from (5.4), we deduce that

$$(1 - \delta - r^\theta \kappa_\theta)\|f\|_{L^\phi(\mathbb{R}^n)} \le \|(f(x_j))_{x_j \in X}\|_{\ell^\phi_w(X)}.$$

In order to prove the right-hand side of the sampling inequality, we consider

$$\begin{aligned}\sum_{j=1}^{m}\phi\left(\frac{|f(x_j)|}{\lambda}\right)\|\beta_j\|_{L^1(\mathbb{R}^n)} &= \sum_{j=1}^{m}\phi\left(\frac{|f(x_j)|}{\lambda}\right)\int_{\mathbb{R}^n}\beta_j(x)dx \\ &\le \sum_{j=1}^{m}\int_{B(x_j;r)}\phi\left(\frac{|f(x)|+|\omega_r(f)(x)|}{\lambda}\right)\beta_j(x)dx \\ &\le \int_{\mathbb{R}^n}\phi\left(\frac{|f(x)|+|\omega_r(f)(x)|}{\lambda}\right)dx \\ &\le \frac{1}{2}\int_{\mathbb{R}^n}\left[\phi\left(\frac{2|f(x)|}{\lambda}\right)+\phi\left(\frac{2|\omega_r(f)(x)|}{\lambda}\right)\right]dx,\end{aligned}$$

Assume that

$$\begin{aligned}\lambda_1 &:= \inf\left\{\tilde{\lambda}>0 : \int_{\mathbb{R}^n}\phi\left(\frac{2|f(x)|}{\tilde{\lambda}}\right)dx \le 1\right\} \\ \lambda_2 &:= \inf\left\{\tilde{\lambda}>0 : \int_{\mathbb{R}^n}\phi\left(\frac{2\omega_r(f)(x)}{\tilde{\lambda}}\right)dx \le 1\right\}.\end{aligned}$$

On substituting $\lambda = (\lambda_1+\lambda_2)$ and proceeding as in Theorem 4.3, we get

$$\begin{aligned}\|(f(x_j))_{x_j\in S}\|_{\ell_w^\phi(S)} &\le 2(\|f\|_{L^\phi(\mathbb{R}^n)}+\|\omega_r(f)\|_{L^\phi(\mathbb{R}^n)}) \\ &\le 2(1+r^\theta\kappa_\theta)\|f\|_{L^\phi(\mathbb{R}^n)}.\end{aligned}$$

This establishes the result. □

From the above result, we conclude that the sample set in C_R with the covering radius satisfying (5.2) is a stable sampling set for the class of δ-concentrated signals in C_R. In the following, we consider i.i.d. random samples X uniformly distributed over C_R and calculate the probability for which X is a stable sample set for $V(R,\delta)$. In view of Theorem 5.1, the probability estimation for the stable sample set reduces to the probability of the event $\{r_X(m)\le\rho\}$ where $0<\rho<1$.

Theorem 5.2. *Let $\phi\in\Delta_2$, $0<\delta<1$ and T be an idempotent integral operator on $L^\phi(\mathbb{R}^n)$ with the kernel K that satisfies* (5.1). *If $X=\{x_j : 1\le j\le m\}$ is a sequence of i.i.d. random variables that are uniformly distributed over C_R then for $\epsilon\in(0,1-\delta)$, the sampling inequality*

$$\begin{aligned}(1-\delta-\epsilon)\|f\|_{L^\phi(\mathbb{R}^n)} &\le \|(f(x_j))_{x_j\in X}\|_{\ell_w^\phi(X)} \\ &\le 2(1+\epsilon)\|f\|_{L^\phi(\mathbb{R}^n)} \quad \forall\, f\in V(R,\delta),\end{aligned} \tag{5.5}$$

holds with the probability at least

$$1-\left[4^nR^n\epsilon^{-n/\theta}\kappa_\theta^{n/\theta}\exp\left(-m\frac{\epsilon^{n/\theta}}{2^nR^n\kappa_\theta^{n/\theta}}\right)\right].$$

Here the weight $w = (\|\beta_j\|_{L^1(\mathbb{R}^n)})_{j=1}^m$ *and* $\{\beta_j : 1 \le j \le m\}$ *is a BUPU of* C_R *associated with some open cover in the neighborhood of random sample points* x_j*.*

Proof. Let $X = \{x_j : 1 \le j \le m\}$ be a set of random points uniformly distributed over C_R. In order to determine the probability estimation for the sampling inequality (5.5), we first need to calculate the probability of the event $\mathcal{E} = \{r_X(m) > \rho\}$. The idea is inspired by the proof [4, Theorem 4.2].

Assume that the event $\mathcal{E}$ is true for some $0 < \rho < 1$ and $\Gamma \subseteq C_R$ is a finite $\rho/2$-net of C_R with the cardinality at most $(4R/\rho)^n$ (see [11, Theorem 5.3]). Then the collection $U = \{B(\gamma; \rho/2) : \gamma \in \Gamma\}$ is an open cover for C_R. Since $r_X(m) > \rho$, there must exists an open set in U which does not contain any sample point, i.e. there exists $\bar{\gamma} \in \Gamma$ such that $B(\bar{\gamma}; \rho/2) \cap X = \emptyset$. Since $\{x_j\}_{j=1}^m$ are independent and identically distributed random points uniformly chosen from C_R, the number of x_j's in any ball is a binomial random variable. As a result, the probability that a specific open set in U contains no points of X is $(1 - (\frac{\rho}{2R})^n)^m$. Therefore, the probability of the event $\mathcal{E}$ is given by

$$\begin{aligned}
P(\mathcal{E}) &= P(r_X(m) > \rho) \\
&\le P(B(\gamma; \rho/2) \cap X = \emptyset \text{ for some } \gamma \in \Gamma) \\
&\le \sum_{\gamma \in \Gamma} P(B(\gamma; \rho/2) \cap X = \emptyset) \\
&\le \left(\frac{4R}{\rho}\right)^n \left(1 - \left(\frac{\rho}{2R}\right)^n\right)^m \\
&\le \left(\frac{4R}{\rho}\right)^n \exp\left(-m \left(\frac{\rho}{2R}\right)^n\right).
\end{aligned}$$

Indeed, Theorem 5.1 implies that the sampling inequality (5.5) holds when the covering radius $r_X(m) \le \rho$, where $\rho = \epsilon^{1/\theta}\kappa_\theta^{-1/\theta}$. Hence the sampling inequality (5.5) holds with probability at least

$$1 - \left[4^n R^n \epsilon^{-n/\theta} \kappa_\theta^{n/\theta} \exp\left(-m \frac{\epsilon^{n/\theta}}{2^n R^n \kappa_\theta^{n/\theta}}\right)\right].$$

This completes the proof. □

Remark 5.1. It can be observed from Theorem 5.2 that the random sampling inequality 5.5 holds with probability $1 - \zeta$ if

$$m > \frac{2^n R^n \kappa_\theta^{n/\theta}}{\epsilon^{n/\theta}} \ln\left(\frac{4^n R^n \epsilon^{-n/\theta} \kappa_\theta^{n/\theta}}{\zeta}\right).$$

For example, the sample size is at least $\mathcal{O}(R^n \ln(R^n))$ for large value of R.

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ORCID

Dhiraj Patel https://orcid.org/0000-0002-4583-3896

S. Sivananthan https://orcid.org/0000-0002-9038-2721

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